Q-learning for CliffBoxPushing

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1. Task, algorithm and controlled experiment

The agent must push a box through the fixed 6×14 map to (4,13). The state is (agent row, agent column, box row, box column); four actions give 7,056 Cartesian states and 28,224 Q entries. We keep the supplied map, reward and step() unchanged. Episodes terminate at a cliff, the goal or 100 steps. Reward is −1 − Manhattan(A,B) − Manhattan(B,G), with −1000 for a cliff and +1000 for reaching the goal.

Q-learning: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma(1-d)\max_b Q(s',b) - Q(s,a)]$. Here d marks termination, so terminal transitions do not bootstrap. Initialize Q to zero; use $\alpha=0.1$, $\gamma=0.99$ and fixed $\varepsilon=0.1$ with uniform greedy tie-breaking. With probability p, replace the intended action uniformly with one of the other three. Update Q for the **intended** action, learning the noisy decision channel.

Train 30,000 episodes per policy for $p_{train} \in \{0,0.1,0.2\}$ and seeds {7,17,27}: nine policies. Evaluate each for 300 episodes per p_{test} with $\varepsilon=0$ and separate RNGs (8,100 evaluation episodes). **Success means box at goal**, not return >0. We use no planning, reward engineering or extra state features.

2. Learning performance

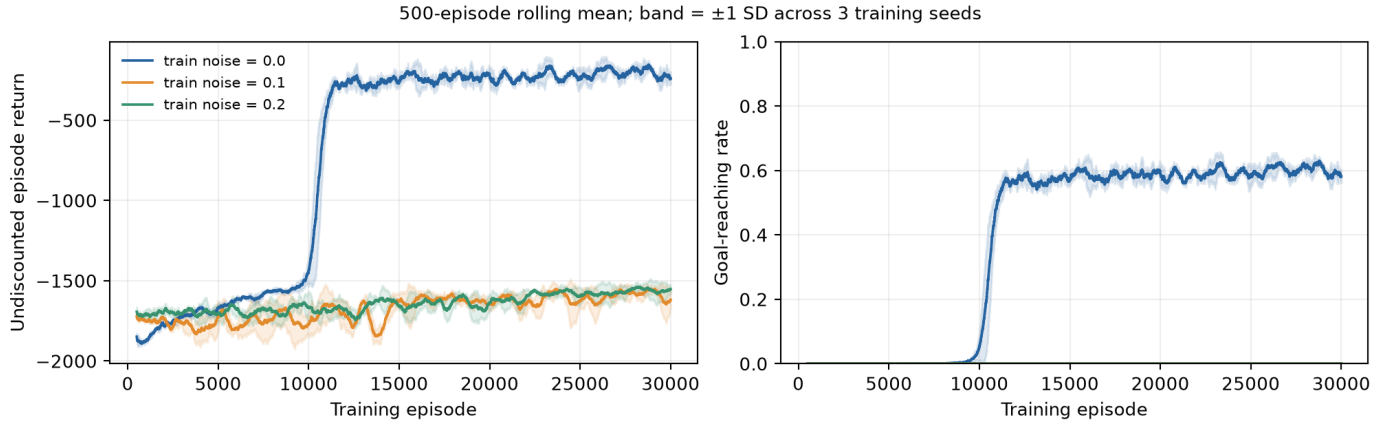


Figure 1. 500-episode moving means; bands are ±1 sample SD across seeds. Clean-trained greedy success is 100.0% (SD 0.0 percentage points), versus 0.0% for 1,000 random episodes (mean return -1403.6). The last 2,000 clean-training episodes succeed 59.7% of the time: ε exploration remains active during training, so training and greedy evaluation differ.

3. A learned clean solution

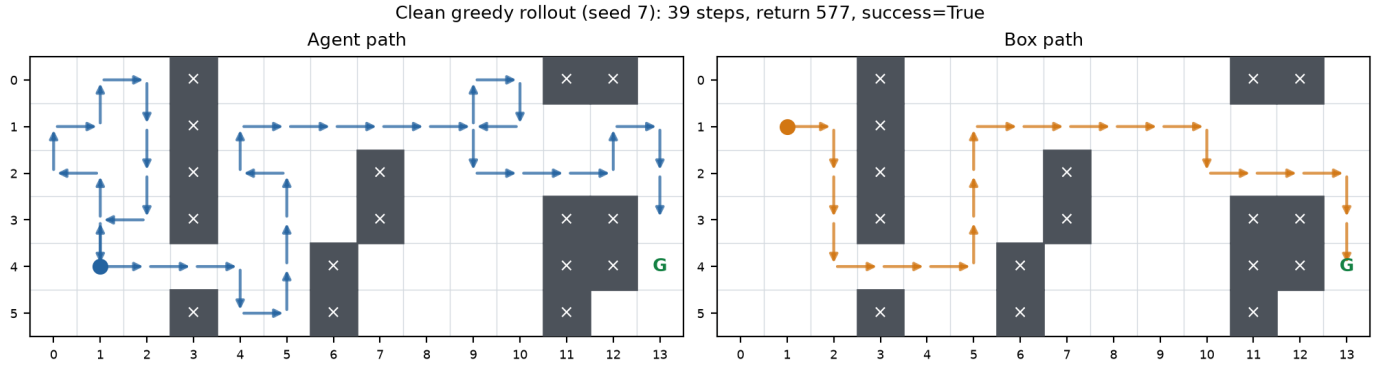


Figure 2. Preselected seed 7, clean-trained greedy rollout: 39 actions, undiscounted return 577, box at goal=True. Blue/orange arrows show the agent/box routes. The agent must reposition to push around cliffs; distance to the goal alone is insufficient.

Robustness and learned policy

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4. Action-noise robustness

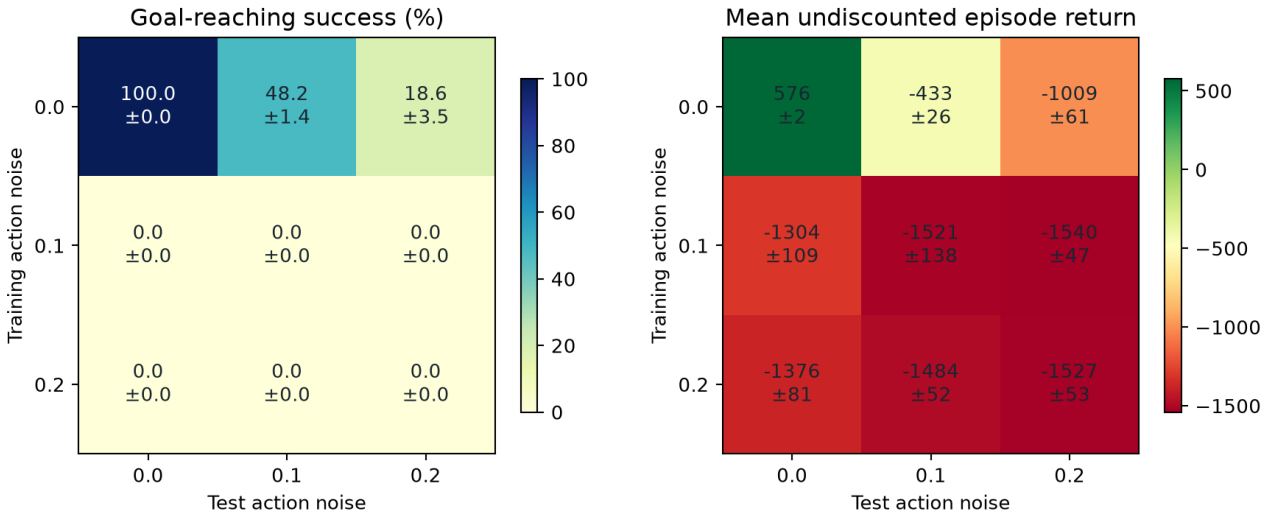


Figure 3. Each cell uses 300 evaluation episodes per seed (900 total). Entries are means ± sample SD across the three training seeds; SD is not a confidence interval. Rows are training noise, columns test noise. No ε exploration or learning is used in evaluation.

For clean training, success falls from 100.0% at test noise 0 to 48.2% at 0.1 and 18.6% at 0.2. A single corrupted action near a cliff can terminate a long trajectory. Training at p=0.1 reduces success at test p=0.1 by 48.2 percentage points; the change at test p=0.2 is -18.6 points. This is a descriptive result under a fixed training budget, not evidence that noise training universally helps. Larger training noise also increases unsafe transitions and makes useful successful experience rarer. **Neither noisy-training setting reached the goal in its 90,000 training episodes across seeds; all their clean greedy evaluations timed out.** This points to unsuccessful goal discovery under these settings, not a successfully learned robust strategy.

5. Values, policy and scope

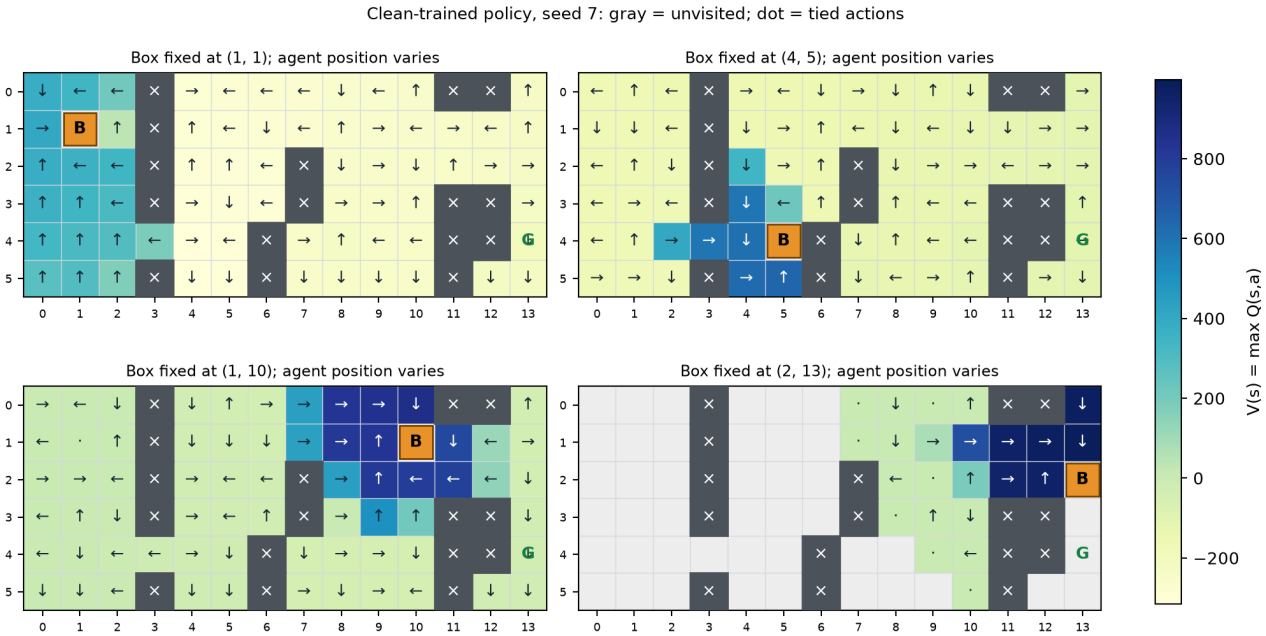


Figure 4. Final clean-trained seed-7 table. Color is $V(s)=\max_a Q(s,a)$ (not divided by state count); arrows are unique greedy actions. Each panel fixes the box and varies the agent position. Gray means unvisited and a dot means a tie. The notebook also plots the entire joint-state table and lists the 20 highest-valued visited states. Unvisited actions can retain optimistic zeros.

Limitations. Three seeds provide limited precision. The four-coordinate state omits time remaining, so the stationary table approximates the 100-step finite-horizon task. Constant α and ε do not prove convergence or route optimality. We retain failed runs and do not attempt the optional UCB/sparse-reward bonuses.

References. Watkins & Dayan (1992), “Q-learning”, Machine Learning 8:279–292. Sutton & Barto (2018), Reinforcement Learning: An Introduction, 2nd ed., §6.5. AI6101 assignment brief and supplied RLAgent_2026Fall.ipynb.